

Real on one dataset.
Gone on the next.

A negative result. Fully open. Every number checkable.

swipe →

THE QUESTION

**Can a model trained on text
actually track a real
human heartbeat —
or just look like it can?**

Text-SAE features over a frozen GPT-2,
tested against real ECG-derived HRV.

THE CONTROLS

(why you can trust what comes next)

- Numbers only — the model never sees the words rest or stress
- Subject-grouped CV — a person is never in train and test
- 1000× permutation null
- Must beat raw-activation, PCA and random baselines

IT LOOKED REAL

WESAD — significant under the strictest split

split	SAE	raw GPT-2	PCA
GroupKFold	0.73	0.78	0.76
leave-1-subj-out	0.72	0.78	0.77

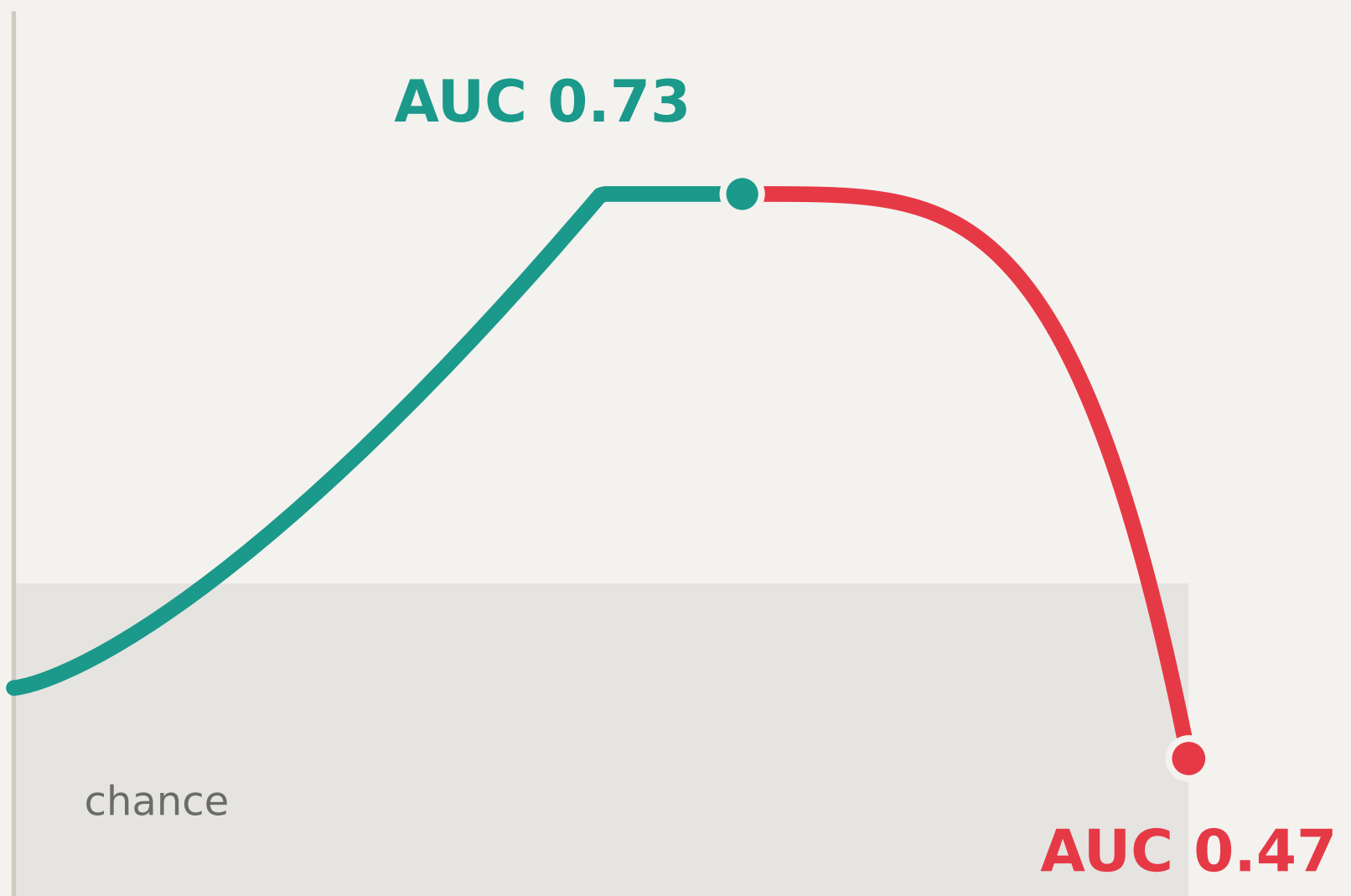
p = 0.001

yes, it was significant —

but SAE never beat the baselines.

IT DID NOT REPLICATE

independent dataset · Stress-Predict · 34 subjects



$p \approx 0.91$ · a cohort artifact, not a finding

THE POINT

**A result that doesn't survive
an independent dataset
isn't a result.**

(the “hidden concept” idea was false too — $p \approx 1.0$)

Code · data · every number:

github.com/ss1738/physiomind-phase2-replication